

Chapter Overview

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18.3.1 Creating a Topic file

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18.1 Visual Basic Help

Help file is a file that is used in place of the default help of VB for the applications. It contains the required help or information for the users. The help file is normally displayed when the user clicks F1 button.

The following three steps are needed to create a help file.

1. Contents File
2. Topic File
3. Project File

18.2 Contents File

The Contents file takes care of the layout and appearance of Contents tab that displays all Headings and Topics of help file. Contents file can be created with **help compiler**.

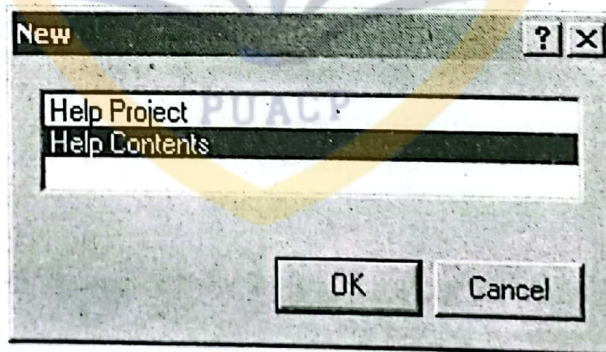
18.2.1 Creating a Contents File

The help software is available in Visual Basic CD (\Common\Tools\VB\HCW). The steps required to develop help file are as follows:

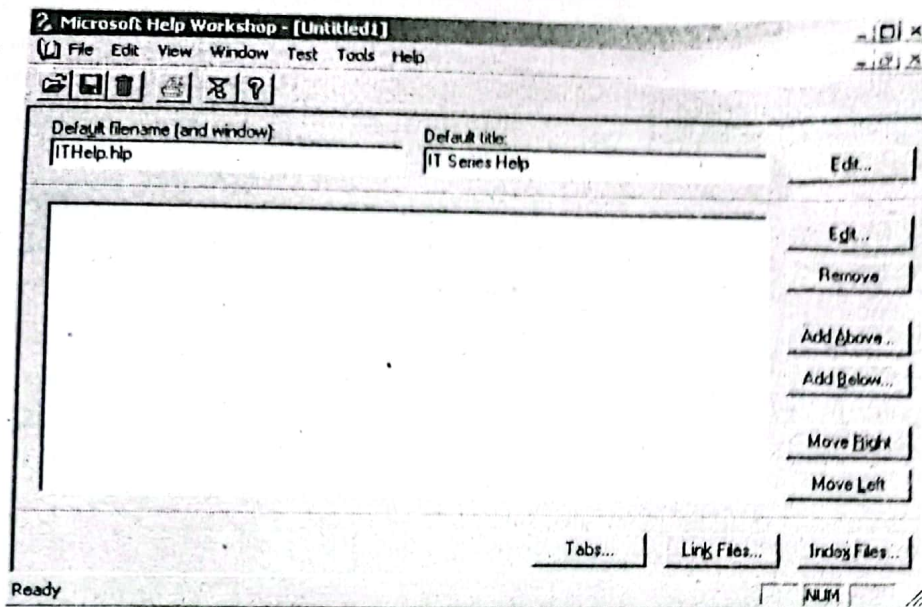
1. Start Help Workshop.



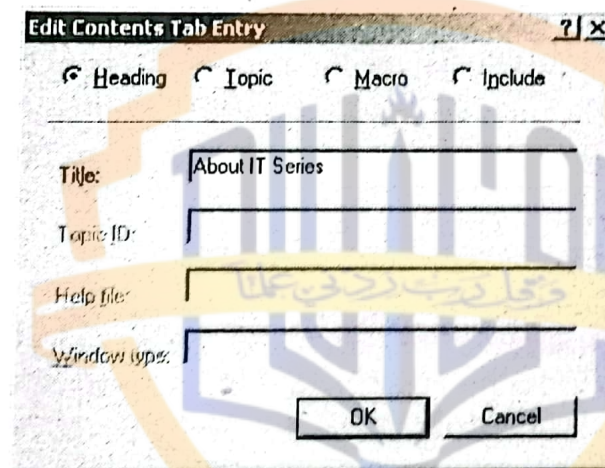
2. Select New from File menu option. The New dialog will appear.



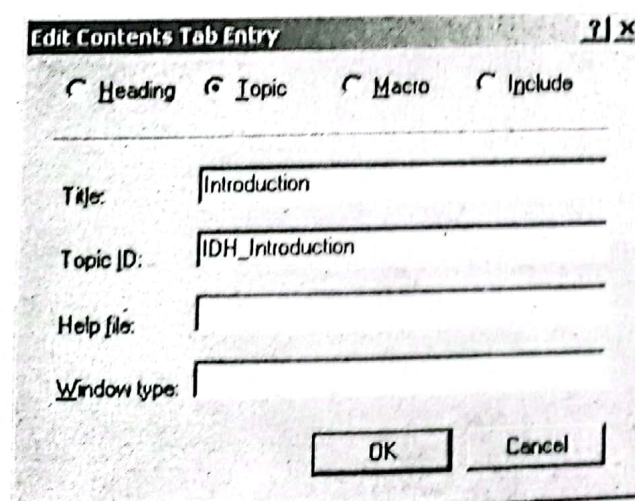
3. Select Help Contents option from New dialog.
4. Type "ITHelp.hlp" as filename for help file in Default filename textbox.
5. Type "IT Series Help" as title for help file in Default title textbox. The Headings and Topics of Help file are created in the same way as menus.
6. Click on Add Above button to create the first heading of help file. This button is used to add a Heading or Topic.



7. The Edit Contents Tab Entry dialog is opened.
8. Select **Heading** option as **Topics** are to be included under some heading.
9. Type the title of the heading as "About IT Series".
10. Click OK.

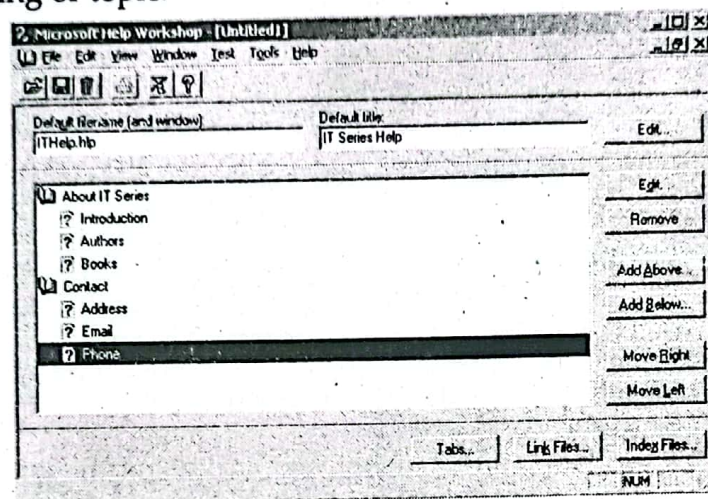


11. Click on Add Below button. The Edit Contents Tab Entry dialog will appear.
12. Select **Topics** option.
13. Type "Introduction" in Title box and "IDH_Introduction" in Topic ID box.
14. Click OK.



15. **Add Below** button is used to add headings and topics. The topic is indented under the heading created before. To add a sub header, click on **Move Right** and add the header. This makes the header the sub header of the previous header.
16. Save your contents file.

The **Edit** button is used to edit heading or topic created already. The **Remove** button is used to delete a heading or topic.

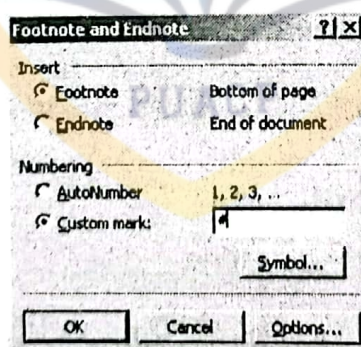


18.3 Topic File

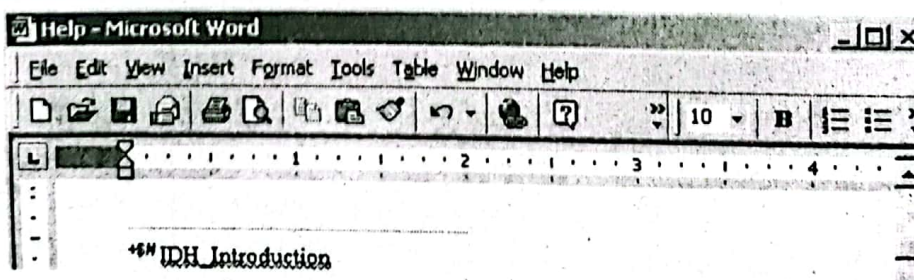
Topic file is a text file that contains the information for help file. You can create a topic file in any word processor that supports footnotes. The file should be saved as .rtf.

18.3.1 Creating a Topic File

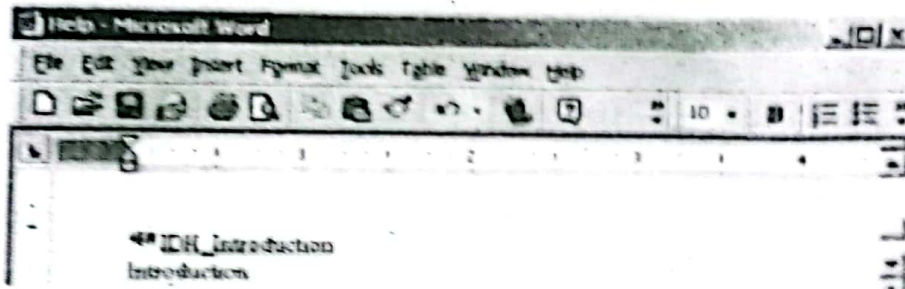
1. Start Microsoft Word.
2. Type the first topic 'Introduction'.
3. Move to the beginning of the topic.
4. Click on **Footnote** from **Insert** menu.
5. Insert the symbol # in the Text box next to the **Custom** mark option.



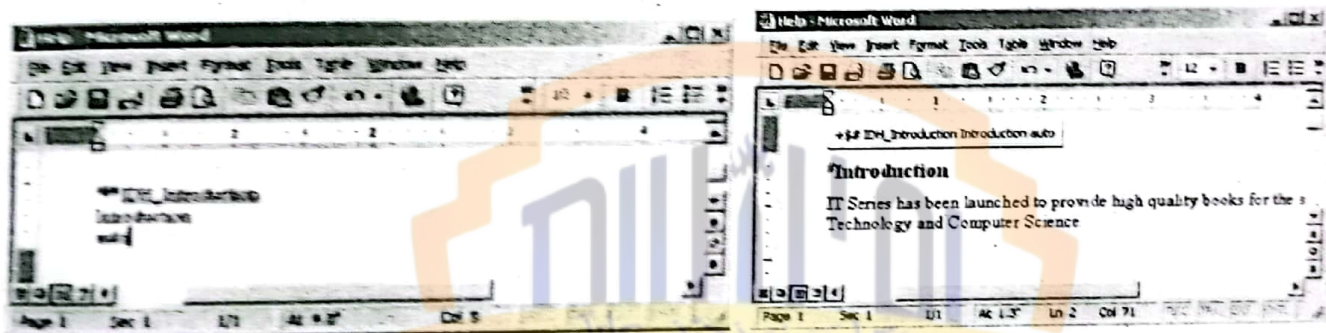
6. Click **OK**.
7. Type the topic ID. This identifies the text specified as the text for that topic.
8. Press **Enter**. The footer will be shown as follows:



9. Insert the second footnote immediately after the first footnote.
10. Insert \$ as Custom mark symbol. This specifies that the text that follows is the title of the topic.
11. Type the title of the topic as the text for the footnote.
12. Press Enter. The Footer would be shown as follows:



13. Insert the third footnote immediately after the second footnote.
14. Insert + as the Custom symbol.
15. Type auto as the text for the footnote. This specifies the help compiler to use an auto browsing sequence.
16. Press Enter. The Footer would be shown as follows:



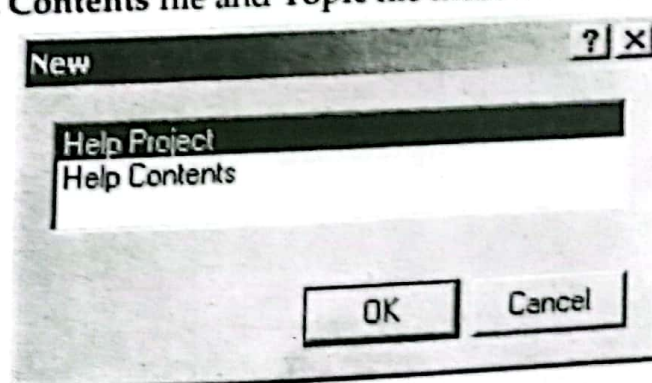
17. Press Ctrl+Enter to insert a new page to enter the next topic. This will insert a Hard Page Break between the topics.

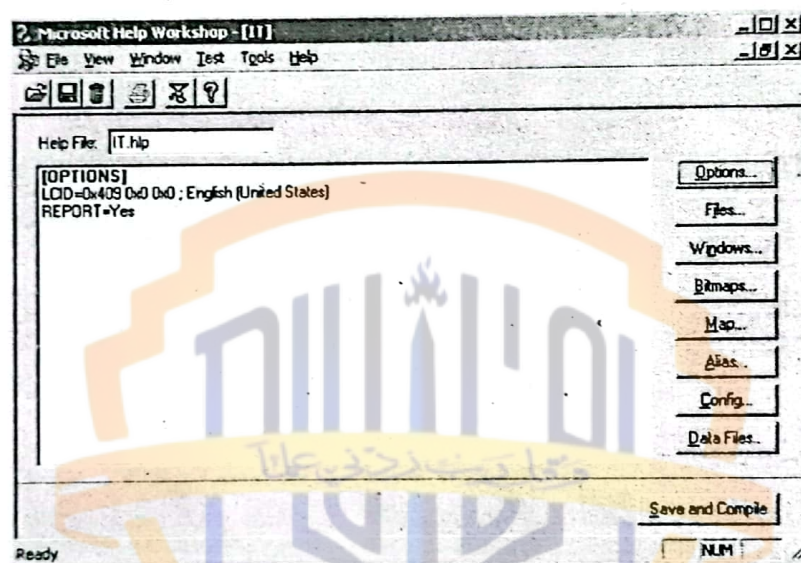
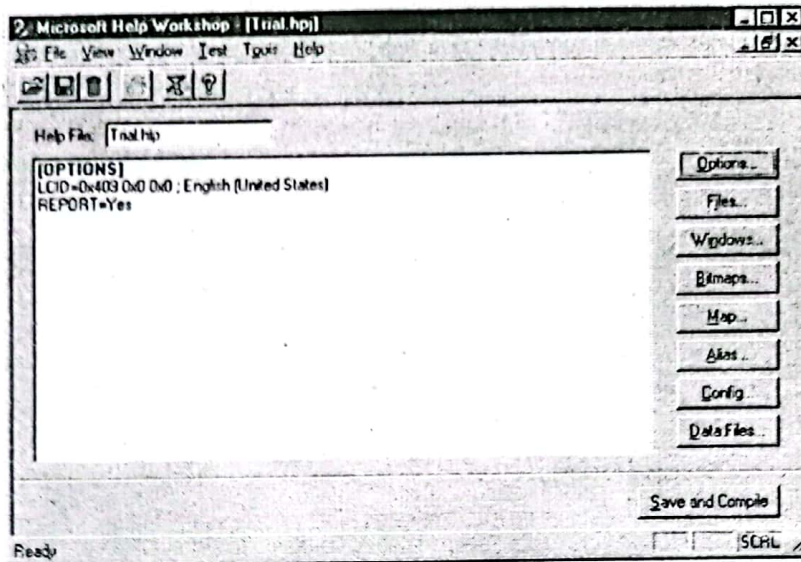
18.4 Project File

The Project file specifies options and links the topic file to Contents file for compiling. It also links other files, such as bitmaps etc. As the Contents file the Project file also needs a help compiler to be created.

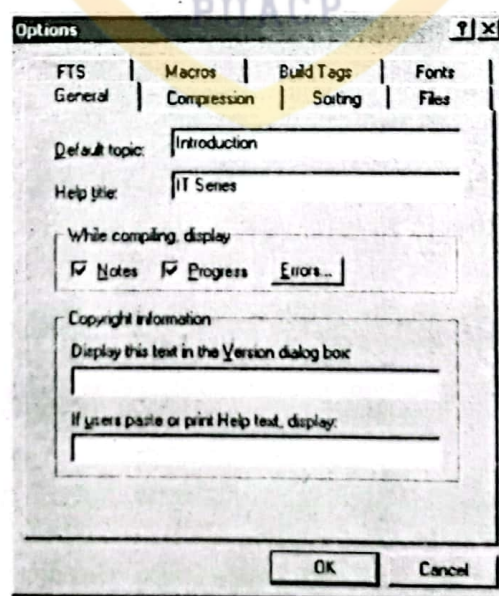
18.4.1 Creating Project Files

1. Select New from File menu from of Help Workshop
2. Click on Help Project. It will open Microsoft Help workshop window. The Project file, Contents file and Topic file must me saved in the same directory.



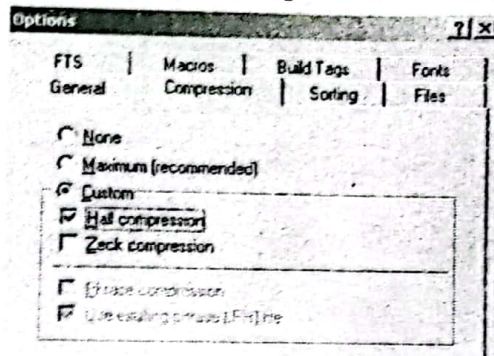


3. Click Options button.
4. Type "Introduction" and "IT Series" in Default topic and Help Title text boxes.

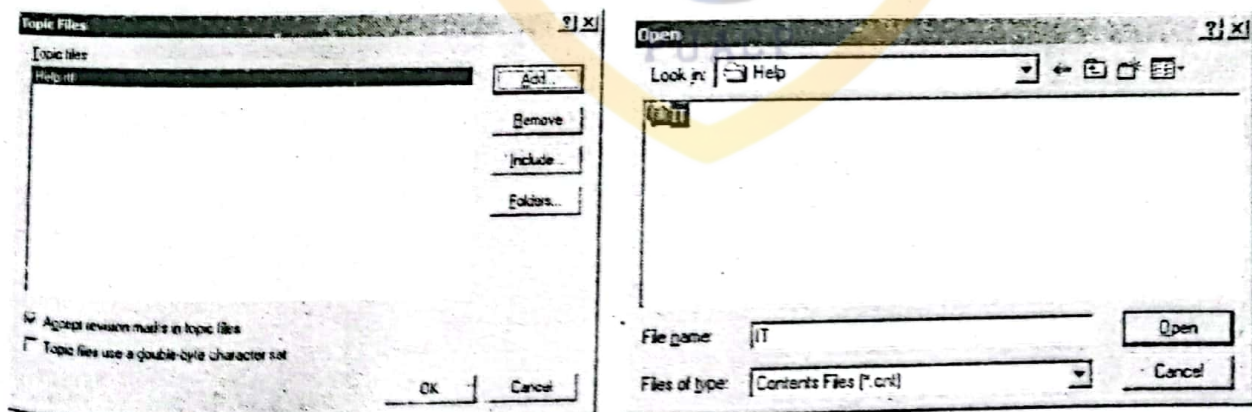
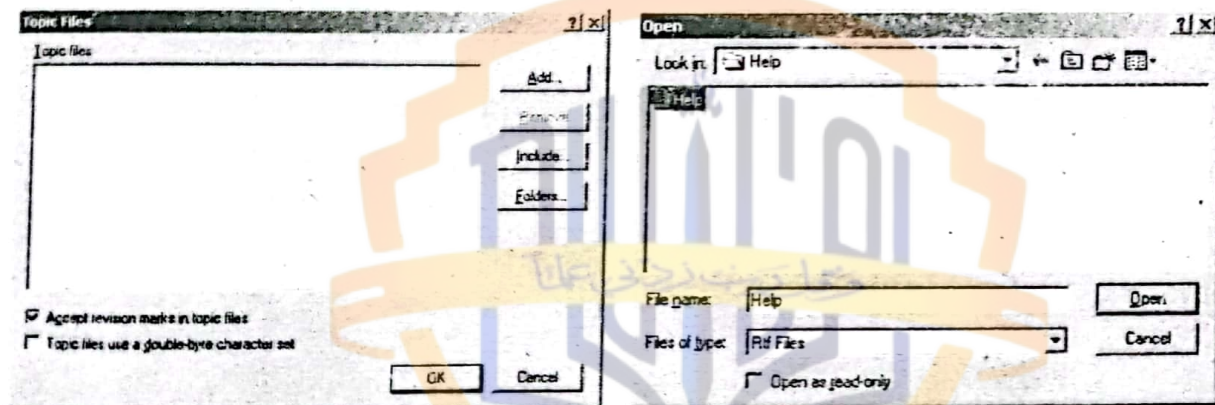


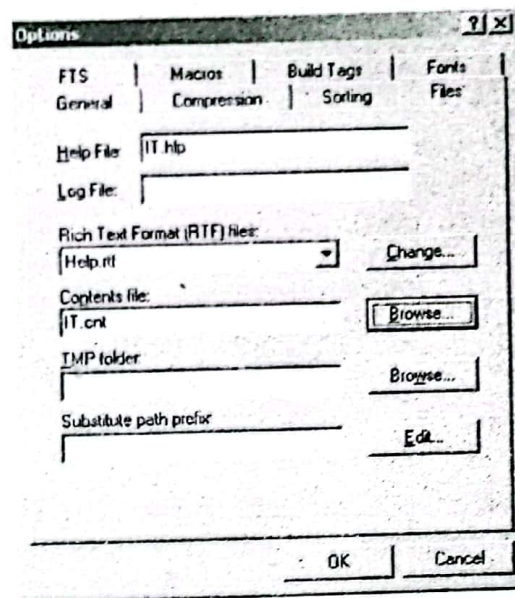
5. Click Compression tab.

6. Click on custom and select Hall compression as follows:

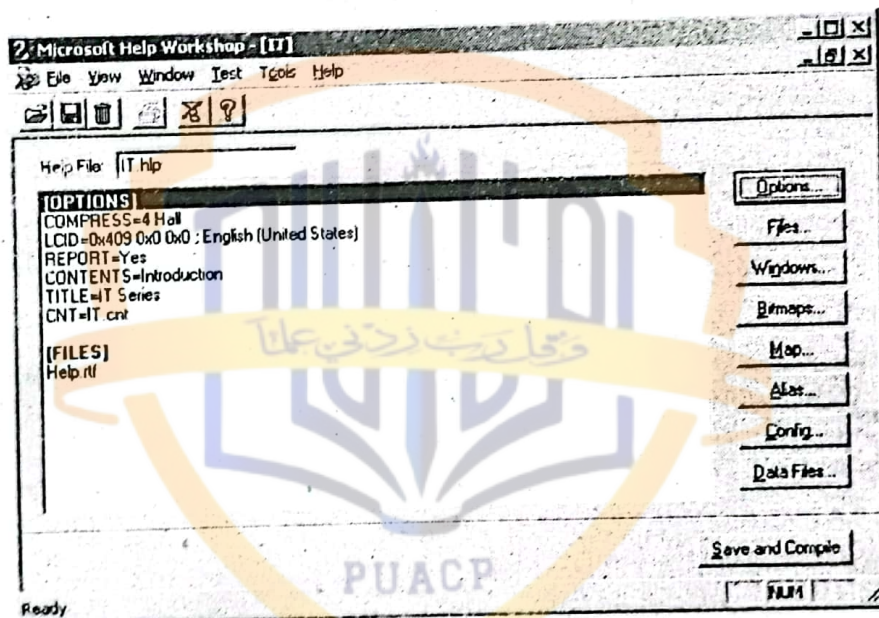


7. Click Files tab.
8. Type the name of the help file in "Help File" text box.
9. Click on Change button next to RTF files field and select the topic file (*.rtf) by clicking on Add button.
10. Click OK.
11. Click on Browse button next to Contents file field and select contents file(*.cnt) and click OK as follows:

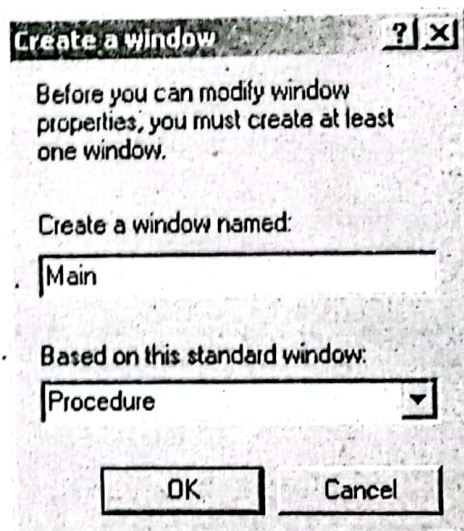




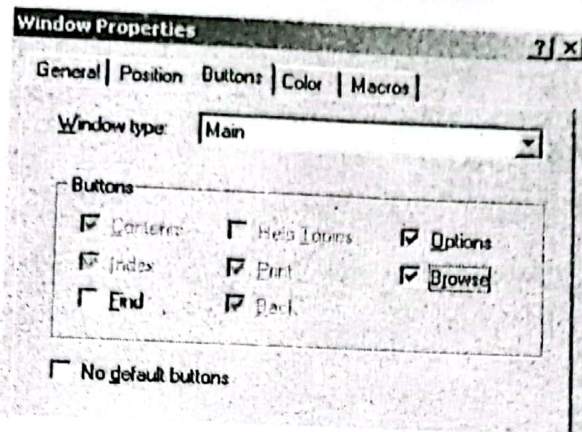
12. The Microsoft Help Workshop window would look as follows:



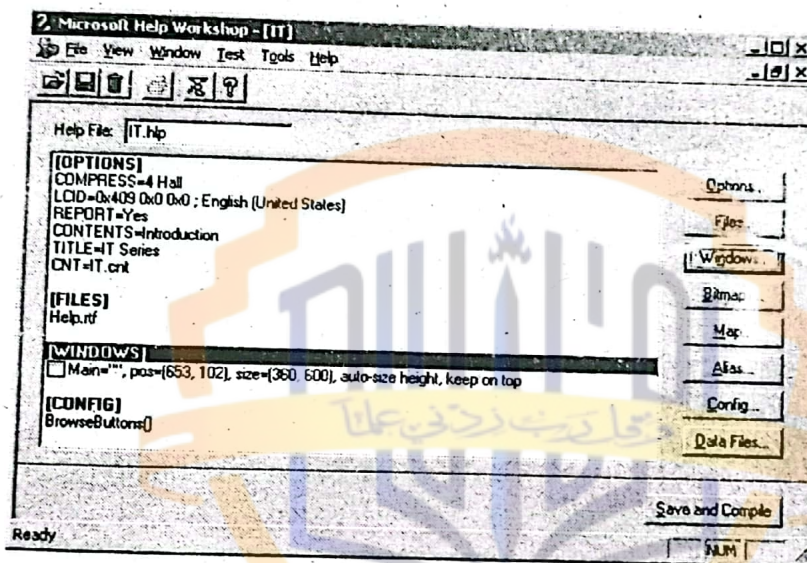
13. Click on Windows button. When Create window prompt appears, type "Main" as window's name and select Procedure in Standard window.



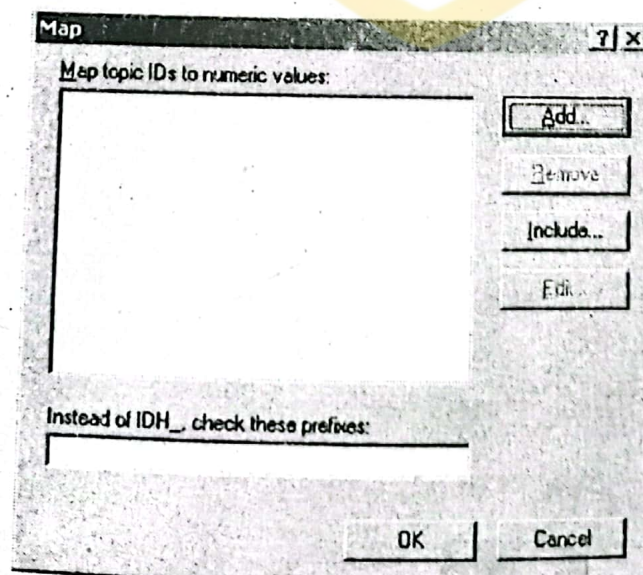
14. In Windows Properties dialog, click on Buttons tab and select Browse.



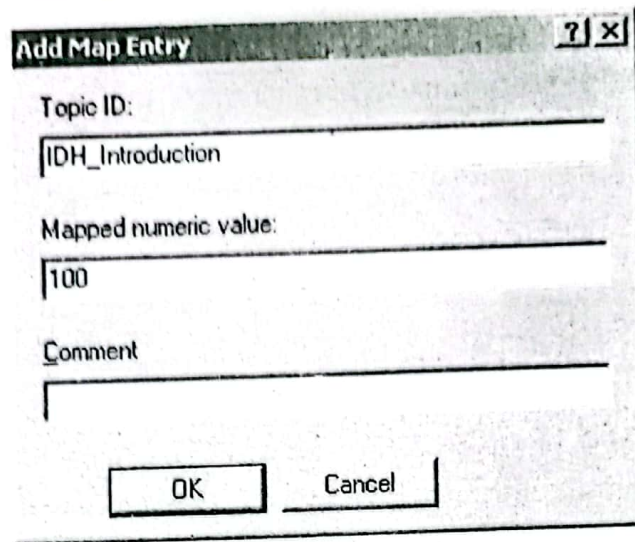
15. This adds browse buttons to help window. Click OK. Microsoft Help Workshop Window would look as below:



16. Click Map button. This option is used to call different topics individually from places in your application.
 17. Click Add button. Enter the first topic ID in the Topic ID field.



18. Enter 100 in Mapped numeric value field. Click OK.



Add Map Entry ? | X |

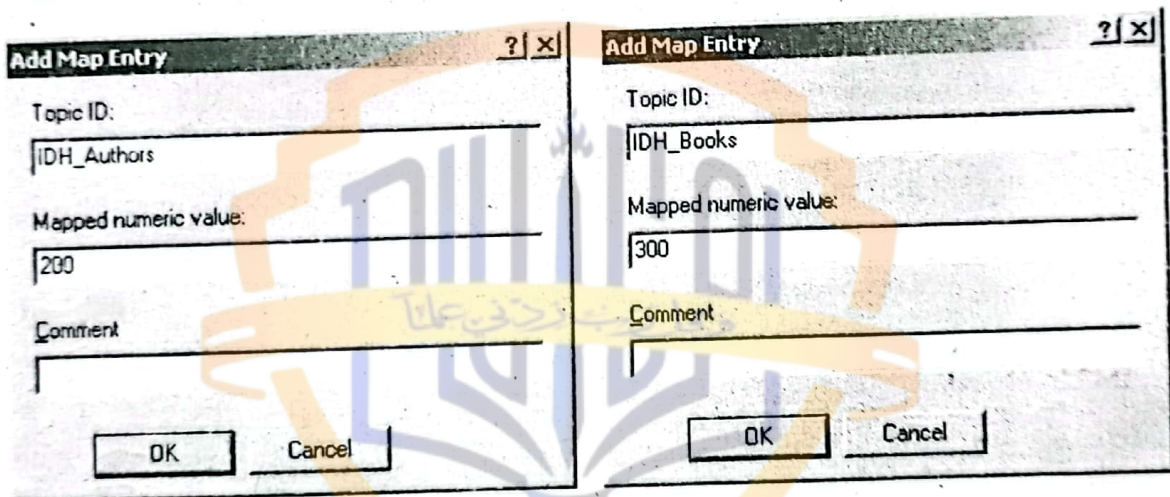
Topic ID:
IDH_Introduction

Mapped numeric value:
100

Comment

OK Cancel

19. Click **Add** button again and enter the second topic ID in **Topic ID** field.
20. Enter 200 in **Mapped numeric value** field.
21. Click **OK** and similarly add other topics.



Add Map Entry ? | X |

Topic ID:
IDH_Authors

Mapped numeric value:
200

Comment

OK Cancel

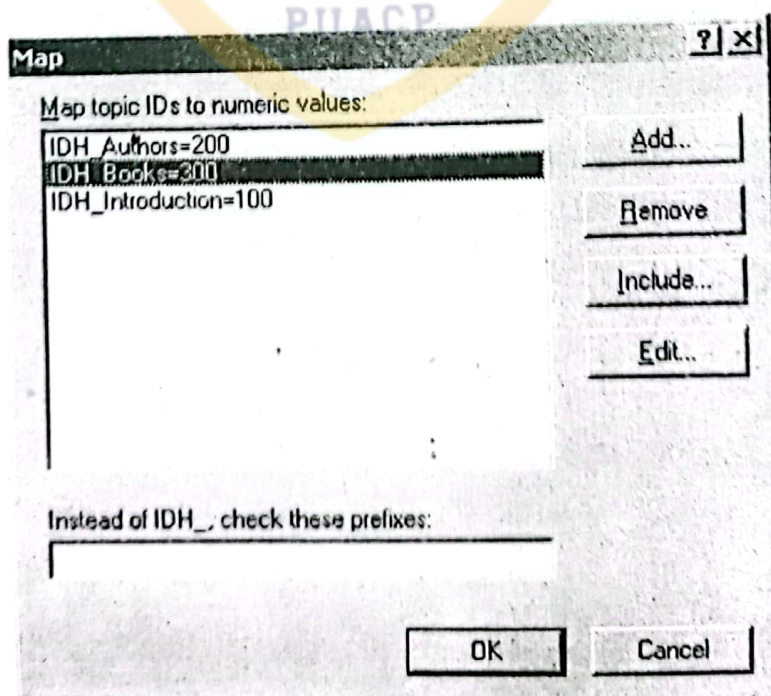
Add Map Entry ? | X |

Topic ID:
IDH_Books

Mapped numeric value:
300

Comment

OK Cancel



Map ? | X |

Map topic IDs to numeric values:

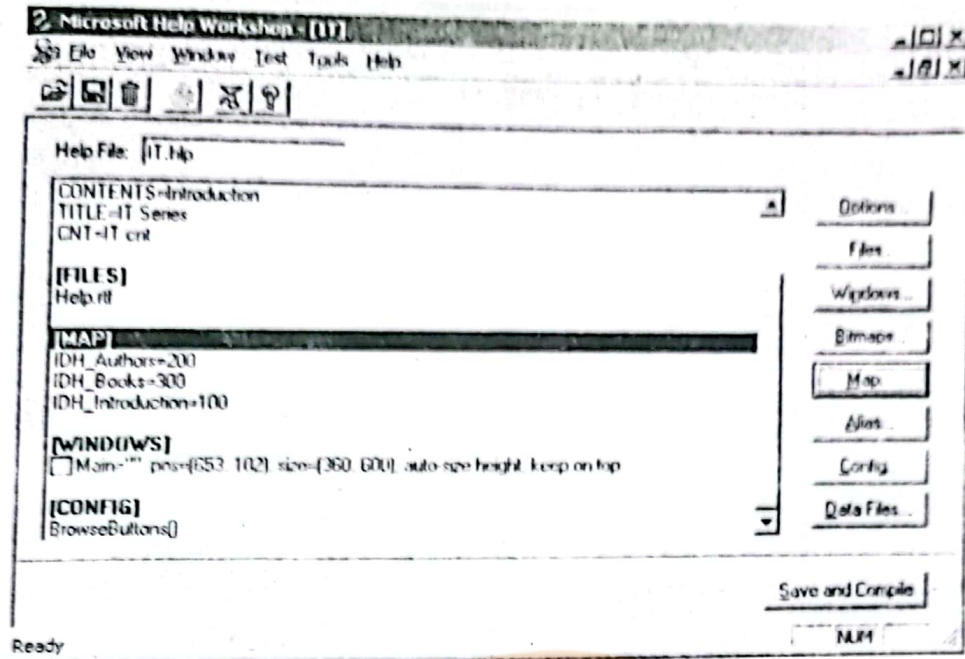
IDH_Authors=200
IDH_Books=300
IDH_Introduction=100

Add..
Remove
Include..
Edit..

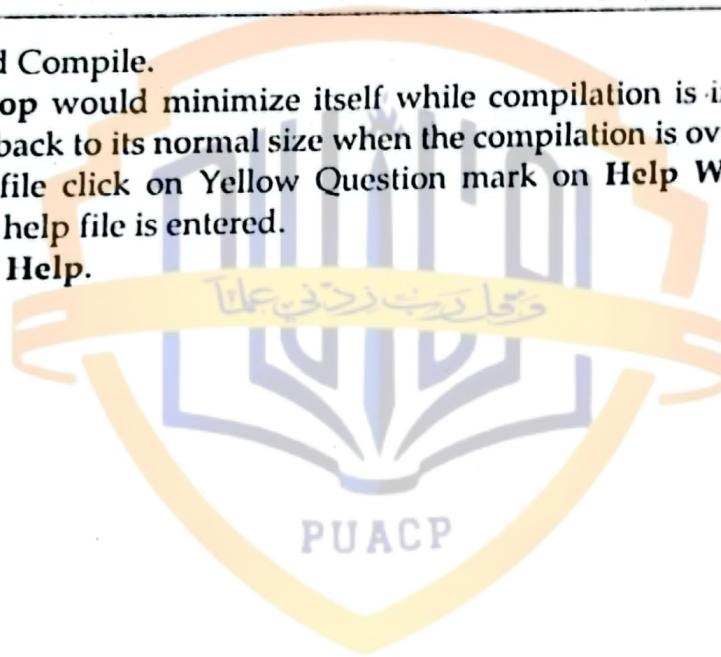
Instead of IDH_, check these prefixes:

OK Cancel

22. The Microsoft Help workshop window looks as follows:



23. Click Save and Compile.
24. **Help Workshop** would minimize itself while compilation is in progress and it would return back to its normal size when the compilation is over.
25. To test **Help** file click on Yellow Question mark on **Help Workshop** toolbar. Make sure the help file is entered.
26. Click on **View Help**.



Programming Exercise Solutions

Chapter 2

Solution 1:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in Click event of the button:
Text1.Text = "Why are you clicking button?"
3. Run the project and check the result.

Solution 2:

1. Start a new Standard EXE project and place three buttons on the form.
2. Set the Captions of the buttons to "1", "2" and "3" respectively.
3. Add the following code in Click event of the first button:
Me.Cls
Print "One"
4. Add the following code in Click event of the second button:
Me.Cls
Print "Two"
5. Add the following code in Click event of the third button:
Me.Cls
Print "Three"
6. Run the project and check the result.

Solution 3:

1. Start a new Standard EXE project and place a textbox, a label and a button on form.
2. Add the following code in Click event of the first button:
Label1.Caption = Text1.Text
3. Run the project and check the result.

Chapter 3

Solution 1:

1. Start a new Standard EXE project.
2. Add the following code in MouseMove event of the form:
Me.Caption = "X-axis: " & X & " Y-axis: " & Y
3. Run the project and check the result.

Solution 2:

1. Start a new Standard EXE project. The default form Form1 will appear.
2. Select Add Form from Project menu to add another form.
3. Add the following code in DbClick event of Form1:
Form1.Hide
Form2.Show
4. Add the following code in DbClick event of Form2:
Form2.Hide
Form1.Show
5. Run the project and check the result.

Solution 3:

1. Start a new Standard EXE project.
2. Place a label on the form.
3. Add the following code in **KeyPress** event of the form:
Label1.Caption = "You pressed a key"
4. Run the project and check the result.

Chapter 4

Solution 1:

1. Start a new Standard EXE project and place three textboxes and a button on the form.
2. Set captions of Label1 to "Enter a number", caption of Label2 to "It's square is" and caption of Label3 to "It's cube is".
3. Add the following code in **Click** event of the button:

```
Dim n As Long
n = Text1.Text
Text2.Text = n * n
Text3.Text = n * n * n
```
4. Run the project and check the result.

Solution 2:

1. Start a new Standard EXE project and place three textboxes and a button on the form.
2. Add the following code in **Click** event of the button:

```
Dim a As Integer, b As Integer, c As Integer
Dim av As Double
a = Text1.Text
b = Text2.Text
c = Text3.Text
av = (a + b + c) / 3
MsgBox "Average age is " & av
```
3. Run the project and check the result.

Solution 3:

1. Start a new Standard EXE project and place a textbox and a button on the form.
2. Add the following code in **Click** event of the button:

```
Dim age As Integer, m As Integer, d As Integer
age = Text1.Text
m = age * 12
d = age * 365
MsgBox "Your age in months is " & m
MsgBox "You age in days is " & d
```
3. Run the project and check the result.

Solution 4:

1. Start a new Standard EXE project and place two textboxes and a button on the form.
2. Add the following code in **Click** event of the button:

```
Const BPRICE = 100
Const PPRICE = 5
Dim Books As Integer, Pencils As Integer
Dim Cost As Integer
Books = Text1.Text
```

```
Pencils = Text2.Text
Cost = Books * BPRICE + Pencils * PPRICE
MsgBox "Total cost is Rs. " & Cost
```

3. Run the project and check the result.

Solution 5:

1. Start a new Standard EXE project and place a textbox and a button on the form.
2. Add the following code in Click event of the button:

```
Dim n As Integer, r As Integer
Dim a As Integer, b As Integer, c As Integer
n = Text1.Text
a = n \ 100
n = n Mod 100
b = n \ 10
n = n Mod 10
MsgBox "Number's reverse order is " & n & b & a
```
3. Run the project and check the result.

Solution 6:

1. Start a new Standard EXE project and place two textboxes and a button on the form.
2. Add the following code in Click event of the button:

```
Dim Temp As String
Temp = Text1.Text
Text1.Text = Text2.Text
Text2.Text = Temp
```
3. Run the project and check the result.

Solution 7:

1. Start a new Standard EXE project and place a textbox and a button on the form.
2. Add the following code in Click event of the button:

```
Dim f As Double, c As Double
f = Text1.Text
c = (f - 32) * 5 / 9
MsgBox "Degree temperature is " & c
```
3. Run the project and check the result.

Solution 8:

1. Start a new Standard EXE project and place a textbox and a button on the form.
2. Add the following code in Click event of the button:

```
Dim c As Double, f As Double
c = Text1.Text
f = 9 / 5 * c + 32
MsgBox "Forenheight temperature is " & f
```
3. Run the project and check the result.

Solution 9:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in Click event of the button:

```
Dim x As Integer, y As Integer, z As Integer
```

```

x = Val(InputBox(" Enter first number", "First number", "0"))
y = Val(InputBox("Enter Second Number", "Second number", "0"))
z = Val(InputBox("Enter Third Number", "Third number", "0"))
Print "Old values : "
Print "x = " & x & " y = " & y & "z = " & z
x = x + y + z
y = x - (y + z)
z = x - (y + z)
x = x - (y + z)
Print " New values : "
Print "x = " & x & " y = " & y & "z = " & z

```

3. Run the project and check the result.

Solution 10:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in Click event of the button:


```

Dim basic As Integer, gross As Single
basic = Val(InputBox("Enter your Basic Salary: ", "Salary", "0"))
gross = basic + (0.35 * basic) + (0.25 * basic)
MsgBox "Your Gross Salary is:" & gross

```
3. Run the project and check the result.

Solution 11:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in Click event of the button:


```

Dim TM As Integer, OM As Integer
Dim Per As Single
OM = Val(InputBox("Enter obtained Marks : "))
TM = Val(InputBox("Enter Total Marks : "))
Per = (OM / TM) * 100
MsgBox "Percentage of the Marks is: " & Per

```
3. Run the project and check the result.

Chapter 5

Solution 1:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in Click event of the button.


```

Dim year As Integer
year = Val(InputBox("Enter year: "))
If (year Mod 4 = 0 And year Mod 100 <> 0) Or (year Mod 400 = 0) Then
  MsgBox (year & " is a leap year.")
Else
  MsgBox (year & " is not a leap year.")
End If

```
3. Run the project and check the result.

Solution 2:

1. Start a new Standard EXE project and place two textboxes and a button on the form.
2. Add the following code in Click event of the button.

```
Dim Math As Integer, Eng As Integer
Math = Text1.Text
Eng = Text2.Text
If Math > 80 And Eng > 80 Then
    MsgBox "Admission granted!"
Elseif Math < 80 And Eng < 80 Then
    MsgBox "Admission rejected!"
Else
    MsgBox "Waiting list..."
End If
```

3. Run the project.

Solution 3:

1. Start a new Standard EXE project and place three textboxes and a button on the form.
2. Add the following code in Click event of the button.

```
Dim a As Integer, b As Integer, c As Integer
a = Text1.Text
b = Text2.Text
c = Text3.Text
If a = b Then
    If a = c Then
        MsgBox "Type of triangle is Equilateral"
    Else
        MsgBox "Type of triangle is Issocoles"
    End If
Elseif a = c Then
    MsgBox "Type of triangle is Issocoles"
Elseif b = c Then
    MsgBox "Type of triangle is Issocoles"
Else
    MsgBox "Type of triangle is Oblique"
End If
```

3. Run the project.

Solution 4:

1. Start a new Standard EXE project and place a textbox and a button on the form.
2. Add the following code in Click event of the button.

```
Dim n As Integer
n = Text1.Text
If n Mod 2 = 0 Then
    MsgBox "The number is even."
Else
    MsgBox "The number is odd."
```

End If

3. Run the project.

Solution 5:

1. Start a new Standard EXE project and place three textboxes and a button on the form.
2. Add the following code in **Click** event of the button.

```
Dim m As Integer, x As Integer, y As Integer, z As Integer
```

```
x = Text1.Text
```

```
y = Text2.Text
```

```
z = Text3.Text
```

```
If x > z Then
```

```
    m = x
```

```
Else
```

```
    m = z
```

```
End If
```

```
Elseif y > z Then
```

```
    m = y
```

```
Else
```

```
    m = z
```

```
End If
```

```
MsgBox "The maximum number is " & m
```

3. Run the project.

Solution 6:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in **Click** event of the button.

```
Dim temp As Single, temp2 As Single, choice As String
```

```
temp = InputBox("Enter temperature: ")
```

```
choice = InputBox("Enter choice: ")
```

```
If choice = "f" Then
```

```
    temp2 = (temp - 32) * 5 / 9
```

```
    Print "Temperature in Celcius is " & temp2
```

```
Elseif choice = "c" Then
```

```
    temp2 = (temp * 9) / 5 + 32
```

```
    Print "Temperature in Fahrenheit is " & temp2
```

```
Else
```

```
    Print "Invalid choice."
```

```
End If
```

3. Run the project.

Solution 7:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in **Click** event of the button.

```
Dim itrate As Integer, salary As Single, itax As Single, nsalary As Single
```

```
nsalary = 0
```

```
salary = InputBox("Enter the salary : ")
```

```
If salary > 30000 Then
```

```

    itax = salary * 20 / 100
    itrate = 20
ElseIf salary >= 20000 Then
    itax = salary * 15 / 100
    itrate = 15
Else
    itax = salary * 10 / 100
    itrate = 10
End If
nsalary = salary - itax
Print "Salary = Rs. " & salary
Print "Your income tax @ " & itrate & "% = Rs. " & itax
Print "Your net salary = Rs. " & nsalary

```

3. Run the project.

Solution 8:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in Click event of the button.
 Dim op As String, side As Integer, base As Single, height As Single, area As Single
 op = InputBox("Enter choice (S for Square, T for Triangle)")
 If op = "S" Or op = "s" Then
 side = Val(InputBox("Enter side: "))
 area = side * side
 Print "Area = " & area
 ElseIf op = "T" Or op = "t" Then
 base = Val(InputBox("Enter base: "))
 height = Val(InputBox("Enter height: "))
 area = base * height * 0.5
 Print "Area = " & area
 End If
3. Run the project.

Solution 9:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in Click event of the button.
 Dim m As Integer
 Dim a As Integer
 m = InputBox("Enter marks: ")
 a = m / 1100 * 100
 If a >= 80 Then
 Print "Percentage: " & a & " Grade: A+"
 ElseIf a >= 70 Then
 Print "Percentage: " & a & " Grade: A"
 ElseIf a >= 60 Then
 Print "Percentage: " & a & " Grade: B"
 ElseIf a >= 50 Then
 Print "Percentage: " & a & " Grade: C"

```

Elseif a >= 40 Then
    Print "Percentage: " & a & " Grade: D"
Elseif a >= 33 Then
    Print "Percentage: " & a & " Grade: E"
Else
    Print "Percentage: " & a & " Grade: F"
End If

```

3. Run the project.

Solution 10:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in Click event of the button.

```

Dim a As Integer, b As Integer, c As Integer
a = Val(InputBox("Enter a number:"))
b = Val(InputBox("Enter a number:"))
c = Val(InputBox("Enter a number:"))
If a = 0 Then
    MsgBox ("Divisor cannot be 0.")
Elseif b Mod a = 0 And c Mod a = 0 Then
    MsgBox a & " is a common divisor of " & b & " and " & c
Else
    MsgBox a & " is not a common divisor of " & b & " and " & c
End If

```

3. Run the project.

Chapter 6

Solution 1:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in Click event of the button.

```

Dim a As Integer, b As Integer, op As Integer
a = InputBox("Enter 1st number:")
b = InputBox("Enter 2nd number:")
op = 1
For i = a To b
    If op = 1 Then
        Print i
        op = 0
    Else
        Print -i
        op = 1
    End If
Next

```

3. Run the project and check the result.

Solution 2:

1. Start a new Standard EXE project and place two textboxes and a button on the form

2. Add the following code in Click event of the button.

```

Dim s1 As String, s2 As String
Dim L As Integer, i As Integer, c As Integer, p As Integer
s1 = Text1.Text
s2 = Text2.Text
L = Len(s1)
i = 1
Do
    p = InStr(i, s1, s2)
    If p > 0 Then
        c = c + 1
        i = p + 1
    Else
        Exit Do
    End If
Loop While (i <= L)
MsgBox "2nd string appears " & c & " times in 1st string."

```

3. Run the project and check the result.

Solution 3:

1. Start a new Standard EXE project and place a textbox and a button on the form.
2. Add the following code in Click event of the button.

```

Dim c As Integer
c = Text1.Text
For i = 1 To 100
    If c Mod 4 = 0 Then
        Print c
    End If
    c = c + 1
Next

```

3. Run the project and check the result.

Solution 4:

1. Start a new Standard EXE project and place a textbox and a button on the form.
2. Add the following code in Click event of the button.

```

Dim n As Integer
n = Text1.Text
For i = 2 To n - 1
    If n Mod i = 0 Then
        MsgBox "The number is not prime!"
        Exit Sub
    End If
Next
MsgBox "The number is prime!"

```

3. Run the project and check the result.

Solution 5:

1. Start a new Standard EXE project and place two textboxes and a button on the form.
2. Add the following code in **Click** event of the button.

```
Dim a As Integer, b As Integer
a = Text1.Text
b = Text2.Text
For i = a To b
    If i Mod 2 = 0 Then
        Print i
    End If
Next
```
3. Run the project and check the result.

Solution 6:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in **Click** event of the button.

```
For i = 1 To 10
    Print i
Next
```
3. Run the project and check the result.

Solution 7:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in **Click** event of the button.

```
For i = 10 To 1 Step -1
    Print i
Next
```
3. Run the project and check the result.

Solution 8:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in **Click** event of the button.

```
For i = 1 To 5
    Print "n= " & i & " Square= " & i * i & " Cube= " & i * i * i
Next
```
3. Run the project and check the result.

Solution 9:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in **Click** event of the button.

```
Dim s As Integer, e As Integer, sum As Integer, op As String
s = InputBox("Enter starting number:")
e = InputBox("Enter ending number:")
op = InputBox("Enter even or odd(e/o):")
If op = "e" Then
    For i = s To e
        If i Mod 2 = 0 Then
```

```

    Print i
    sum = sum + i
End If
Next
MsgBox "The sum of numbers is " & sum
ElseIf op = "o" Then
    For i = s To e
        If i Mod 2 <> 0 Then
            Print i
            sum = sum + i
        End If
    Next
    MsgBox "The sum of numbers is " & sum
Else
    MsgBox "Invalid option"
End If

```

3. Run the project and check the result.

Solution 10:

1. Start a new Standard EXE project place a button on the form.
2. Add the following code in Click event of the button.

```

Dim a As Integer, b As Integer, nxt As Integer, limit As Integer
limit = Val(InputBox("Enter the limit up to which Fibonacci terms required:"))
a = 0
b = 1
Print "Fibonacci terms upto limit are"
Print a
Print b
nxt = a + b
While (nxt <= limit)
    Print nxt
    a = b
    b = nxt
    nxt = a + b
Wend

```
3. Run the project and check the result.

Solution 11:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in Click event of the button.

```

Dim r As Double
r = 1
For n = 1 to 100 Step 2
    r = r + (1/n)
Next

```

Print "Result is " & r

3. Run the project and check the result.

Solution 12:

1. Start a new Standard EXE project and place two textboxes and a button on the form.
2. Add the following code in **Click** event of the button.

```
Dim a As Integer, b As Integer, p As Integer
```

```
a = Text1.Text
```

```
b = Text2.Text
```

```
p = 1
```

```
For i = 1 To b
```

```
    p = p * a
```

```
Next
```

```
Print "Result is " & p
```

3. Run the project and check the result.

Solution 13:

1. Start a new Standard EXE project and a button on the form.

2. Add the following code in **Click** event of the button.

```
Dim a As Integer, b As Integer, nxt As Integer, n As Integer, count As Integer
```

```
n = Val(InputBox("How many fibonnaccl terms required:"))
```

```
a = 0
```

```
b = 1
```

```
Print "Fibonacci terms are" & vbCrLf & a & vbCrLf & b
```

```
count = 2
```

```
Do While count < n
```

```
    nxt = a + b
```

```
    Print nxt
```

```
    count = count + 1
```

```
    a = b
```

```
    b = nxt
```

```
Loop
```

3. Run the project and check the result.

Solution 14:

1. Start a new Standard EXE project and a button on the form.

2. Add the following code in **Click** event of the button.

```
Dim marks As Integer, numberOfStudent As Integer, i As Integer
```

```
Dim marks1 As Double, marks2 As Double
```

```
marks1 = -1
```

```
marks2 = -2
```

```
numberOfStudents = InputBox("Enter the number of students: ")
```

```
For i = 0 To numberOfStudents - 1
```

```
    marks = InputBox("Enter a student marks: ")
```

```
    If marks > marks1 Then
```

```
        marks2 = marks1
```

```

marks1 = marks
Elseif marks > marks2 Then
    marks2 = marks
End If
Next i
Print "Highest marks is " & marks1
Print "Second highest marks is " & marks2

```

3. Run the project and check the result

Solution 15:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in **Click** event of the button.

```

Dim x As Integer, a As Integer, r As Integer, sum As Integer
sum = 0
x = InputBox("Enter an integer Number: ", "Enter Number", "0")
a = x
While (x <> 0)
    r = x Mod 10
    If (r = 0) Then
        sum = sum + x
    Else
        sum = sum + r
        x = x / 10
    End If
Wend
Print "The sum of digits of " & a & " = " & sum;

```

3. Run the project and check the result

Solution 16:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in **Click** event of the button.

```

Dim num As Integer, sum As Integer, min As Integer, max As Integer, count As Integer
Dim avg As Single
sum = 0
count = 0
num = InputBox("Enter positive number: ")
min = num
max = num
Do While num >= 0
    sum = sum + num
    count = count + 1
    If num > max Then
        max = num
    Elseif num < min Then
        min = num
    End If

```

```

num = InputBox("Enter positive number: ")
Loop
If count = 0 Then
    Print "No positive number entered. "
Else
    avg = sum / count
    Print "You entered " & count & " numbers."
    Print "Average = " & avg & vbNewLine
    & "Minimum = " & min & vbNewLine
    & "Maximum = " & max
End If

```

3. Run the project and check the result

Solution 17:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in **Click** event of the button.

```
Dim a As Integer, i As Integer, incre As Integer
```

```

a = 1
incre = 3
i = 1
Do While a <= 40
    Print a:
    a = a + incre
    i = i + 1

```

```
Loop
```

3. Run the project and check the result

Solution 18:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in **Click** event of the button.

```
Dim a As Double, b As Double, c As Double, d As Integer
```

```
d = 0
```

```
a = CDBl(Text1.Text)
```

```
For b = 1 To a
```

```
    c = b * b
```

```
    If c = a Then
```

```
        d = 1
```

```
    End If
```

```
Next b
```

```
If d = 1 Then
```

```
    MsgBox "Perfect Square", vbInformation
```

```
Else
```

```
    MsgBox "Not Perfect Square", vbInformation
```

```
End If
```

3. Run the project and check the result

Solution 19:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in Click event of the button.

```
Dim num1 As Integer, num2 As Integer, lcm As Integer, gcd As Integer
Dim temp As Integer, prod As Integer, n1 As Integer, n2 As Integer
```

```
Do
```

```
    num1 = Val(InputBox("Enter first positive numbers:"))
```

```
    num2 = Val(InputBox("Enter Second Positive Number"))
```

```
Loop While num1 <= 0 Or num2 <= 0
```

```
n1 = num1
```

```
n2 = num2
```

```
prod = n1 * n2
```

```
Do While n2 <> 0
```

```
    If (n1 Mod n2) = 0 Then
```

```
        Exit Do
```

```
    Else
```

```
        Do While n2 <> 0
```

```
            If (n1 Mod n2) = 0 Then
```

```
                Exit Do
```

```
            Else
```

```
                temp = n1 Mod n2
```

```
                n1 = n2
```

```
                n2 = temp
```

```
            End If
```

```
        Loop
```

```
        gcd = n2
```

```
        lcm = prod \ gcd
```

```
        Print "L.CM of " & num1 & " and " & num2 & " = " & lcm
```

```
        Print "GCD of " & num1 & " and " & num2 & " = " & gcd
```

```
    End If
```

```
Loop
```

3. Run the project and check the result

Solution 20:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in Click event of the button.

```
Dim n As Integer, s As Integer, t As Integer, e As Integer
```

```
t = Val(InputBox("Enter table number: "))
```

```
s = Val(InputBox("Enter starting point: "))
```

```
e = Val(InputBox("Enter ending point: "))
```

```
For n = s To e
```

```
    Print t & " * " & n & " = " & t * n
```

```
Next n
```

3. Run the project and check the result

Solution 21:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in Click event of the button.

```
Dim n As Integer, sum As Integer
sum = 0
n = 3
While n <= 99
    sum = sum + n
    n = n + 3
Wend
Print "Sum = " & sum
```
3. Run the project and check the result

Solution 22:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in Click event of the button.

```
Dim number As Integer
Dim factor As Integer
number = Val(InputBox("Enter a positive integer: "))
Print "The factors for " & number & " is "
factor = 2
Do While factor <= number
    If number Mod factor = 0 Then
        number = number \ factor
        Print factor
    Else
        factor = factor + 1
    End If
Loop
```
3. Run the project and check the result

Solution 23:

1. Start a new Standard EXE project and place a button on the form.
2. Add the following code in Click event of the button.

```
Dim i As Integer, n As Integer
Dim p As Integer, j As Integer
n = InputBox("Enter a number: ")
For i = 1 To n
    p = 1
    For j = 2 To i - 1
        If i Mod j = 0 Then
            p = 0
            Exit For
        End If
    Next j
```

```

If p = 1 Then
    Print i
End If

```

```

Next i

```

3. Run the project and check the result

Chapter 7

Solution 1:

1. Start a new Standard EXE project.
2. Place a button on the form.
3. Add the following code in Click event of the button.

```

Dim Names(4) As String
For i = 0 To 4
    Names(i) = InputBox("Enter name:")
Next
For i = 0 To 4
    If InStr(Names(i), "A") = 1 Then
        Print Names(i)
    End If
Next

```

4. Run the project and check the result.

Solution 2:

1. Start a new Standard EXE project.
2. Place a button on the form.
3. Add the following code in Click event of the button.

```

Dim Num(4) As Integer
For i = 0 To 4
    Num(i) = InputBox("Enter No.")
Next
Print "Numbers in original order:"
For i = 0 To 4
    Print Num(i)
Next
Print "Numbers in reverse order:"
For i = 4 To 0 Step -1
    Print Num(i)
Next

```

4. Run the project and check the result.

Solution 3:

1. Start a new Standard EXE project and Place a button on the form.
2. Add the following code in Click event of the button.

```

Dim n(9) As Integer, Max As Integer
For i = 0 To 9
    n(i) = InputBox("Enter No.")

```

```

Next
Max = n(0)
For i = 1 To 9
    If n(i) > Max Then
        Max = n(i)
    End If
Next
Print "Maximum number is " & Max

```

3. Run the project and check the result.

Solution 4:

1. Start a new Standard EXE project and Place a button on the form.
2. Add the following code in Click event of the button.

```

Dim n(9) As Integer, Min As Integer
For i = 0 To 9
    n(i) = InputBox("Enter No.")

```

```

Next
Min = n(0)
For i = 1 To 9
    If n(i) < Min Then
        Min = n(i)
    End If

```

```

Next
Print "Minimum number is " & Min

```

3. Run the project and check the result.

Solution 5:

1. Start a new Standard EXE project and Place a button on the form.
2. Add the following code in Click event of the button.

```

Dim n(4) As Integer, Even As Integer, Odd As Integer
For i = 0 To 4

```

```

    n(i) = InputBox("Enter No.")

```

```

    If n(i) Mod 2 = 0 Then

```

```

        Even = Even + 1

```

```

    Else

```

```

        Odd = Odd + 1

```

```

    End If

```

```

Next

```

```

Print "Total even numbers are " & Even

```

```

Print "Total odd numbers are " & Odd

```

3. Run the project and check the result.

Solution 6:

1. Start a new Standard EXE project and Place a button on the form.
2. Add the following code in Click event of the button.

```

Dim Names(4) As String, n As String

```

```

Names(0) = "Tariq Mahmood"
Names(1) = "Imran Saeed"
Names(2) = "Tasleem Mustafa"
Names(3) = "Ahsan Raza"
Names(4) = "IT Series"
n = InputBox("Enter name to find")
For i = 0 To 4
    If Names(i) = n Then
        MsgBox Names(i) & " found at index " & i
        Exit Sub
    End If
Next
MsgBox n & " not found in the array."

```

3. Run the project and check the result.

Chapter 8

Solution 1:

1. Start a new Standard EXE project.
2. Place a list box, a textbox and a button on the form.
3. Add the following code in Click event of the button.
List1.AddItem Text1.Text
4. Run the project and check the result.

Solution 2:

1. Start a new Standard EXE project.
2. Place two textboxes, four option buttons and a command button on the form.
3. Set the captions of the radio buttons to "Plus", "Minus", "Multiply" and "Divide" respectively.
4. Set the names of the radio buttons to "optPlus", "optMinus", "optMultiply" and "optDivide" respectively.
5. Add the following code in Click event of the button.

```
Dim a As Integer, b As Integer, c As Integer
```

```
a = Text1.Text
```

```
b = Text2.Text
```

```
If optPlus.Value = True Then
```

```
    c = a + b
```

```
Elseif optMinus.Value = True Then
```

```
    c = a - b
```

```
Elseif optMultiply.Value = True Then
```

```
    c = a * b
```

```
Else
```

```
    c = a / b
```

```
End If
```

```
MsgBox "Result is " & c
```

6. Run the project and check the result.

Solution 3:

1. Start a new Standard EXE project and place a combo box, and a button on the form.
2. Add the following code in the **Load** event of the form.
`Combo1.AddItem "Islamabad"`
`Combo1.AddItem "Lahore"`
`Combo1.AddItem "Faisalabad"`
`Combo1.AddItem "Jhang"`
`Combo1.AddItem "Multan"`
3. Add the following code in **Click** event of the button.
`MsgBox Combo1.Text`
4. Run the project and check the result.

Chapter 9

Solution 1:

1. Start a new Standard EXE project and select **Menu Editor** from **Tools** menu.
2. To create **File** menu, set **Caption** to **&File** and **Name** to **mnuFile**.
3. Click **Next**.
4. Click the right arrow (→) button.
5. To create **Open** menu, set **Caption** to **&Open** and **Name** to **mnuOpen**.
6. Click **Next**.
7. Click the right arrow (→) button to create a submenu.
8. Set **Caption** to **&Form**, **Name** to **mnuForm** and **Shortcut** to **Ctrl+F**.
9. Click **Next**.
10. To create **Project** submenu, set **Caption** to **&Project**, **Name** to **mnuProject** and **Shortcut** to **Ctrl+P**.
11. Click **Next**.
12. Click the left arrow (←) button.
13. To create **Close** submenu, set **Caption** to **&Close** and **Name** to **mnuClose**. Disable the menu by unchecking the **Enabled** checkbox.
14. Click **Next**.
15. To create **Exit** submenu, set **Caption** to **&Exit** and **Name** to **mnuExit**.
16. Click **Next**.
17. Click the left arrow (←) button.
18. To create the **Edit** menu, set **Caption** to **&Edit** and **Name** to **mnuEdit**.
19. Click **Next**.
20. Click the right arrow (→) button to add items for the **Edit** menu.
21. To create **Copy** menu item, set **Caption** to **&Copy**, **Name** to **mnuCopy** and **Shortcut** to **Ctrl+C**.
22. Click **Next**.
23. To create **Paste** menu item, set **Caption** to **&Paste**, **Name** to **mnuPaste** and **Shortcut** to **Ctrl+V**.
24. Click **Next**.
25. To insert a divider, set **Caption** to **- (one dash)** and **Name** to **Sep**.

26. Click Next.
27. To create **Search** menu item, set **Caption** to &Search, **Name** to mnuSearch and **Shortcut** to F3.
28. Click Next.
29. To create **Replace** menu item, set **Caption** to &Replace, **Name** to mnuReplace and **Shortcut** to F4.
30. Click Next.
31. To create **Case Sensitive** menu item, set **Caption** to Case Sensitive and **Name** to mnuCase.
32. Click the **Checked** checkbox and click **OK**.

Chapter 10

Solution 1:

1. Start a new Standard EXE project and place a textbox and a button on the form.
2. Add the following in General Declaration section.

```
Sub Table(n As Integer)
    For i = 1 To 10
        Print i & "*" & n & "=" & i * n
    Next
End Sub
```

3. Add the following code in Click event of the button.

```
Dim num As Integer
num = Text1.Text
Call Table(num)
```

4. Run the project and check the result.

Solution 2:

1. Start a new Standard EXE project and place a textbox and a button on the form.
2. Add the following in General Declaration section.

```
Function Fact(n As Integer) As Long
    Dim f As Long
    f = 1
    For i = 1 To n
        f = f * i
    Next
    Fact = f
End Function
```

3. Add the following code in Click event of the button.

```
Dim num As Integer, Result As Long
num = Text1.Text
Result = Fact(num)
MsgBox "Factorial is " & Result
```

4. Run the project and check the result.

Solution 3:

1. Start a new Standard EXE project and Place two textboxes and a button on the form.

26. Click Next.
27. To create **Search** menu item, set **Caption** to &Search, **Name** to mnuSearch and **Shortcut** to F3.
28. Click Next.
29. To create **Replace** menu item, set **Caption** to &Replace, **Name** to mnuReplace and **Shortcut** to F4.
30. Click Next.
31. To create **Case Sensitive** menu item, set **Caption** to Case Sensitive and **Name** to mnuCase.
32. Click the **Checked** checkbox and click **OK**.

Chapter 10

Solution 1:

1. Start a new Standard EXE project and place a textbox and a button on the form.
2. Add the following in General Declaration section.

```
Sub Table(n As Integer)
    For i = 1 To 10
        Print i & " * " & n & " = " & i * n
    Next
End Sub
```

3. Add the following code in Click event of the button.

```
Dim num As Integer
num = Text1.Text
Call Table(num)
```

4. Run the project and check the result.

Solution 2:

1. Start a new Standard EXE project and place a textbox and a button on the form.
2. Add the following in General Declaration section.

```
Function Fact(n As Integer) As Long
    Dim f As Long
    f = 1
    For i = 1 To n
        f = f * i
    Next
    Fact = f
End Function
```

3. Add the following code in Click event of the button.

```
Dim num As Integer, Result As Long
num = Text1.Text
Result = Fact(num)
MsgBox "Factorial is " & Result
```

4. Run the project and check the result.

Solution 3:

1. Start a new Standard EXE project and Place two textboxes and a button on the form.